

Prep for Session

- **Open GitHub to your profile (e.g., github.com/lcjohnso)**
 - **Create an account if you don't already have one!**
- **My Tools (not required, but could be useful)**
 - **GitHub Desktop (<https://desktop.github.com/>)**
 - **VS Code for Editor**
- **Files on REU GitHub (<https://bit.ly/cierareu-github>)**
 - **Download GitTutorial_commands.txt file**

A Brief Introduction to git and GitHub



git



GitHub

Cliff Johnson
Northwestern University and Adler Planetarium



Alts: Subversion/svn, Mercurial

Version Control



Alts: BitBucket, GitLab

Repository Hosting Service
repository = “repo”

"FINAL".doc



FINAL.doc!



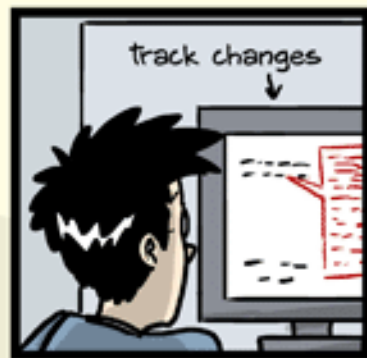
FINAL_rev.2.doc



FINAL_rev.6.COMMENTS.doc



FINAL_rev.8.comments5.
CORRECTIONS.doc



FINAL_rev.18.comments7.
corrections9.MORE.30.doc



FINAL_rev.22.comments49.
corrections.10.#@\$%WHYDID
ICOMETOGRADSCHOOL?????.doc



Goals

- Keep track of changes to files
- Revert back to old versions of files
- Test changes to code/files without losing original
- Synchronizes code between different people, facilitating collaborative work

git commands:

add

commit

push

```
git add <file>
```

```
git commit -m  
"<descriptive message>"
```

**Note: [blah] is optional, <blah> means fill in with your own text,
<<blah>> is a comment you should omit**


```
mkdir ghtutorial-2026  
cd ghtutorial-2026  
echo "# README: ghtutorial-2026" >> README.md  
  
git init  
  
git add README.md  
git commit -m "first commit"
```



```
git status
```

```
git log [--oneline]
```

git push origin master

Remote Name

Remote Branch

```
git remote add origin https://github.com/<user>/<repo>.git  
git push --set-upstream origin main
```

But first: initialize repo on GitHub through browser

git clone <repo url>

```
git clone https://github.com/<user>/<repo>.git
```

```
git clone https://github.com/CIERA-Northwestern/CIERA_REU.git
```

Or use GitHub Desktop

git reset

git reset --hard <hash>

git reset --soft <hash>

git reset --hard HEAD

(git revert)

git branch

```
git branch <new_branch>
```

```
git checkout <new_branch>
```

```
git checkout -b <new_branch>
```

```
git branch -D <branch>
```

`git merge`

Handling a dreaded merge conflict!

Tangents

Markdown

- See Cheat Sheet = <https://www.markdownguide.org/cheat-sheet/>

Github Security Best Practices

- DON'T COMMIT PASSWORDS, KEYS, OR CREDENTIALS

Github Authentication

- Optional: 2FA and personal access tokens
- Desktop based OSX Keychain usage = <https://desktop.github.com/>
- `git config --global credential.helper 'cache --timeout=3600'`

GitHub Tutorial: Part 2

- Open source and collaborative workflows
 - Forks & Pull Requests
 - Development Patterns
- Intro to other useful GitHub features
 - Issues
 - GitHub Pages

Open Source Code

- Using other people's code
 - If published to PyPI: `pip install <package>`
 - If not published to PyPI: clone/download + install

GitHub Collaboration

- Development Install: instead of installing a particular version of code/library, install
 - Use case: to stay up-to-date with latest code changes (between releases), to make local custom changes
 - From code directory (which contains setup.py file):

```
pip install -e .
```

GitHub Collaboration

- Two Methods:
 - Create a branch
 - Fork a repo

GitHub Collaboration

- Make a change, submit a pull request
 - Make changes on local copy of fork
 - Push to remote copy of fork
- Use GitHub interface to create pull request to transfer changes back to original repo.

GitHub Collaboration

- Staying Up-to-date
 - Setup addition “upstream” remote

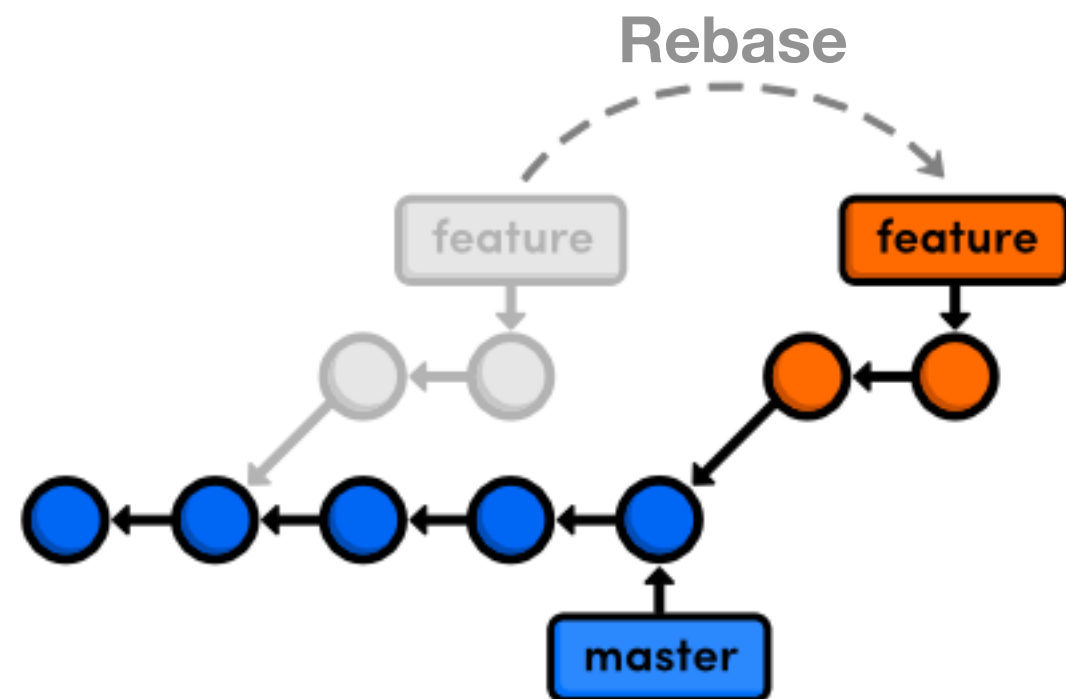
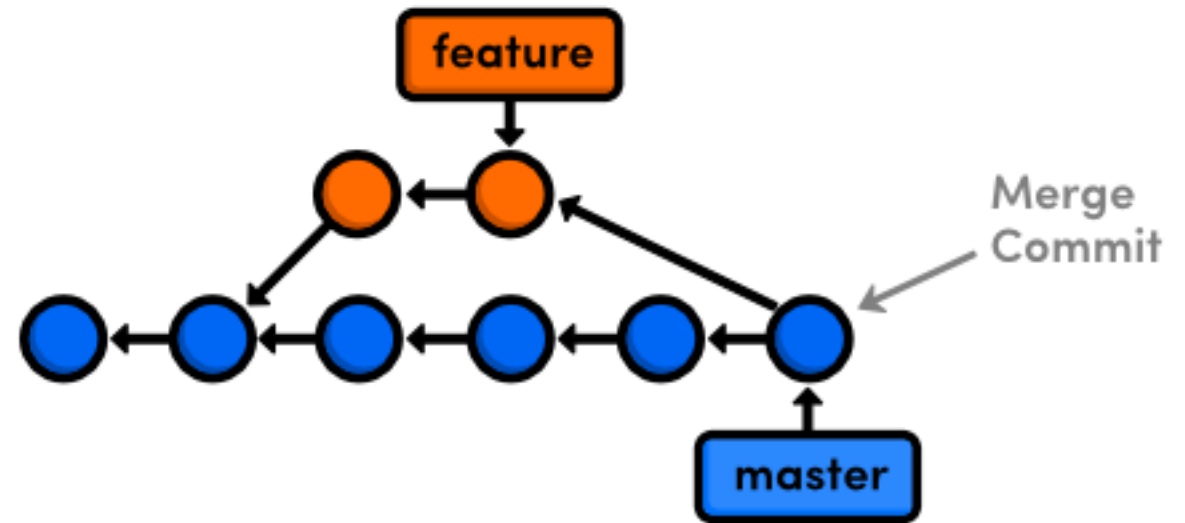
```
git remote add upstream <repo_url>
```

- Sync with upstream repo

```
git fetch upstream master  
git merge upstream/master
```

GitHub Collaboration

- Rebase vs. Merge



GitHub Pages

- GitHub provides free hosting!
- Visit the “Settings” page for a repo, and select from available hosting options: host contents of ``main`` branch or ``/docs`` folder

Sources

An Introduction to Git and GitHub by Brian Yu

Link = https://www.youtube.com/watch?v=MJUJ4wbFm_A

NU EECS-330 Assignment Doc

Link = <https://docs.google.com/document/d/17bt57nFiLo60ehKhSid5lSFofiXCXY5xKqZ2PSZb5g4/edit>

Git Tower FAQ

<https://www.git-tower.com/learn/git/faq>

GitHub - Adding Existing Projects

<https://help.github.com/en/github/importing-your-projects-to-github/adding-an-existing-project-to-github-using-the-command-line>

Git Rebasing

<https://code.tutsplus.com/tutorials/rewriting-history-with-git-rebase--cms-23191>